Hydrogen or Single Electron Atoms

We start by recalling a few facts about single-electron atoms in the absence of the corrections to be considered below. We are concerned with atoms with Z protons and a single electron, so that the potential is given by[1]

$$V(r) = -\frac{Z\alpha}{r}, \tag{1}$$

where $\alpha = e^2/4\pi \simeq 1/137$ is the fine structure constant. This is a central potential problem and the solutions to Schrödinger's equation can be factorized into a radial and an angular part,

$$\psi(r, \theta, \phi) = R_{n,l}(\rho) Y_{l,m}(\theta, \phi). \tag{2}$$

where $R_{n,l}(\rho)$ is the radial wavefunction and $Y_{l,m}(\theta, \phi)$ are the spherical harmonics. We also defined

$$\rho = \frac{2r}{na_0}, \tag{3}$$

a dimensionless radial variable with respect to the Bohr radius a_0,

$$\boxed{a_0 = \frac{1}{mZ\alpha}}. \tag{4}$$

For the (unperturbed) ground state of the atom, a_0 is the most probable distance between the electron and the nucleus. Note that the atomic number will appear always alongside α as $Z\alpha$. Up to finite nuclear size corrections, we can think of this combination as the effective coupling constant of the electromagnetic interactions in the atom.

> The stronger the EM interactons, the larger α is, and the smaller the Bohr radius becomes. Similarly, the heavier the electron, the smaller the Bohr radius becomes. In essence, the Bohr radius is determined by the balance between kinetic and potential energies. Increasing α (or $Z\alpha$) *increases* the strength of the potential, which pulls the electron closer to the nucleus. On the other hand, increasing m *decreases* the kinetic energy for a given momentum ($T = p^2/2m$), which also pulls the electron closer to the nucleus.

Recall that the spherical harmonics $Y_{l,m}(\theta, \phi)$, which are eigenfunctions of L^2 and L_z with eigenvalues $\hbar^2 l(l+1)$ and $\hbar m$, respectively. The angular part is normalized, $\int d\Omega |Y_{l,m}(\theta, \phi)|^2 = 1$. The radial part of the wavefunction is given by the radial equation,

$$\left[\frac{\hat{\mathbf{p}}^2}{2m} + \hat{V}(r)\right] u_{n,l}(r) = E_n^{(0)}\, u_{n,l}(r), \tag{5}$$

where $u_{n,l}(r) = rR_{n,l}(r)$ is the reduced radial wavefunction. On the left-hand side, the first term is the kinetic energy of the electron, the second is the centrifugal barrier, and the last term is the energy associated with our central potential. An equivalent form for the equation is in terms of radial and transverse momentum operators, $\hat{\mathbf{p}} = \hat{p}_r \frac{\hat{\mathbf{r}}}{r} + \hat{\mathbf{p}}_\perp$, where $\hat{\mathbf{p}}_\perp$ is the transverse momentum operator (n.b., the hat means an operator here, not a unit vector). In this language,

$$\left[\frac{\hat{p}_r^2}{2m} + \frac{\hat{\mathbf{p}}_\perp^2}{2m} + \hat{V}(r)\right] u_{n,l}(r) = \left[\frac{\hat{p}_r^2}{2m} + \frac{\hat{\mathbf{L}}^2}{2mr^2} + \hat{V}(r)\right] u_{n,l}(r) = E_n^{(0)} u_{n,l}(r), \tag{6}$$

where $\hat{p}_r = -i\frac{d}{dr}$ and $\hat{\mathbf{p}}_\perp^2 = \hat{\mathbf{L}}^2/r^2$. This form makes is obvious that the electron can only acquire a non-zero momentum in the transverse direction if it has a non-zero angular momentum, since $\hat{\mathbf{p}}_\perp^2 = \hat{\mathbf{L}}^2/r^2$.

[1] From now on, we will work in natural units $\hbar = c = 1$ and set $\epsilon_0 = 1$. See the notes on natural units for more details.

In the position basis,

$$\begin{aligned}
\langle \mathbf{r}| \frac{\hat{\mathbf{p}}^2}{2m} |\psi\rangle &= -\frac{1}{2m}\nabla^2\psi(\mathbf{r}), &(7)\\
\langle \mathbf{r}| \frac{\hat{\mathbf{L}}^2}{2mr^2} |\psi\rangle &= \langle \mathbf{r}| \frac{\hat{\mathbf{p}_\perp}^2}{2m} |\psi\rangle = \frac{1}{2mr^2} l(l+1)\psi(\mathbf{r}), &(8)\\
\langle \mathbf{r}| \hat{V}(\mathbf{r}) |\psi\rangle &= V(r)\psi(\mathbf{r}). &(9)
\end{aligned}$$

The derivation of the radial wavefunction can be found in many standard textbooks. We recall the final result here,

$$R_{n,l}(r) = \left[\left(\frac{2}{na_0}\right)^3 \frac{(n-l-1)!}{2n(n+l)!} \right]^{1/2} e^{-\rho/2}\rho^l L^{2l+1}_{n-l-1}(\rho), \tag{10}$$

where $L^{2l+1}_{n-l-1}(\rho)$ are the associated Laguerre polynomials. For the ground state, $n = 1$ and $l = 0$, we have

$$R_{1,0}(r) = \frac{2}{\sqrt{a_0^3}} e^{-r/a_0}, \tag{11}$$

where we used $L^1_0(\rho) = 1$.

Expectation Values: Working with the Laguerre polynomials in all generality can be quite messy, but we will usually focus on specific cases. In particular, in terms of expectation values, the following are quite standard results:

$$\begin{aligned}
\langle n, l, m|n, l, m\rangle &= \int_0^\infty dr r^2 R_{n,l}(r)^2 = 1, &(12)\\
\langle r^2\rangle_{n,l,m} &= \frac{a_0^2 n^2}{2} \left(5n^2 - 3l(l+1) + 1\right), &(13)\\
\langle r\rangle_{n,l,m} &= \frac{a_0}{2} \left(3n^2 - l(l+1)\right), &(14)\\
\langle r^{-1}\rangle_{n,l,m} &= \frac{1}{n^2 a_0}, &(15)\\
\langle r^{-2}\rangle_{n,l,m} &= \frac{1}{n^3 a_0^2} \frac{1}{(l+1/2)}, &(16)\\
\langle r^{-3}\rangle_{n,l,m} &= \frac{1}{n^3 a_0^3} \frac{1}{l(l+1/2)(l+1)} \quad \text{for } l \neq 0. &(17)
\end{aligned}$$

The last relation accounts for the fact that for sufficiently large exponent k in r^{-k}, the expectation value $\langle r^{-k}\rangle_{n,l,m}$ becomes singular at $r = 0$. Inspecting Eq. (10), we can deduce that we must have $2l+2-k > 0$ (or $l > k/2-1$) for the expectation value to be well-defined. For $k > 3$, the expectation values can be found by recursion relations. We will not dwell on those here.

The unperturbed energy levels are given by

$$\boxed{E^{(0)}_n = -\frac{m(Z\alpha)^2}{2n^2}.} \tag{18}$$

The atomic spectrum is uniquely defined by the quantum numbers

$$n, l, m \quad \text{with } n \geq 1, \quad 0 \leq l \leq n-1, \quad -l \leq m \leq l. \tag{19}$$

Therefore, for any given n, there are

$$\sum_{l=0}^{n-1} (2l+1) = 2\left(\frac{n(n-1)}{2}\right) + n = n^2 \tag{20}$$

energy-degenerate states.

Typical Momentum of the Electron. We can ask what is the most probable momentum of the electron in each orbital. Let us split this question into the radial and transverse components, $\hat{\mathbf{p}} = \hat{p}_r(\mathbf{r}/r) + \hat{\mathbf{p}}_\perp$. For the ground state (Eq. (11)), this is a simple calculation:

$$\langle \hat{p}_r^2 \rangle_{100} = \left(\frac{na_0}{2}\right) \int_0^\infty d\rho \rho^2 e^{-\rho/2} \left(\frac{d^2}{d\rho^2} e^{-\rho/2}\right) = \frac{1}{a_0^2} \quad \Longrightarrow \quad \sqrt{\langle \hat{p}_r^2 \rangle_{100}} = \frac{1}{a_0}. \tag{21}$$

where we used $r^2 dr = (na_0/2)^3 \rho^2 d\rho$, $\hat{p}_r^2 = -\frac{d^2}{dr^2} = (2/na_0)^2 \frac{d^2}{d\rho^2}$. Note that $\langle \hat{p}_r \rangle_{100} = 0$, as expected from the spherical symmetry of the problem. Similarly, we could compute the expectation value of the transverse momentum to find that it is only non-zero for $l \neq 0$. With the help of $L^2 = l(l+1)$ and Eq. (16), we have

$$\langle \hat{\mathbf{p}}_\perp^2 \rangle_{n,l,m} = l(l+1)\langle r^{-2} \rangle_{n,l,m} = \frac{l(l+1)}{n^3 a_0^2 (l+1/2)} \Longrightarrow \sqrt{\langle \hat{\mathbf{p}}_\perp^2 \rangle_{n,l,m}} = \frac{\sqrt{l(l+1)}}{n^{3/2} a_0 \sqrt{l+1/2}}. \tag{22}$$

as expected. Note that for large n, the maximum transverse momentum is decreasing as $1/n$.

Virial Theorem. The virial theorem is a rather generic relation between potential and kinetic energy. For the single-electron atom, it can be extremely useful: for a potential of the form $V(r) \propto r^k$, the expectation value of the kinetic energy $\langle T \rangle$ and potential energy $\langle V \rangle$ are related by

$$\langle T \rangle = \frac{k}{2} \langle V \rangle. \tag{23}$$

In our case, $k = -1$, so we have $\langle T \rangle = -\frac{1}{2}\langle V \rangle$. In particular, the expectation value of the potential is (with the help of Eq. (15)) as

$$\langle V \rangle_{n,l,m} = -\frac{Z\alpha}{n^2 a_0} = -\frac{m(Z\alpha)^2}{n^2} = 2E_n^{(0)}. \tag{24}$$

Note that this had to be the case since by virtue of the virial theorem,

$$E_n^{(0)} = \langle T \rangle + \langle V \rangle = \frac{1}{2} \langle V \rangle. \tag{25}$$

Let us use this result to compute the expectation value of the radial momentum:

$$\langle \hat{p}_r^2 \rangle_{n,l,m} = 2m\langle T \rangle - \langle \hat{\mathbf{p}}_\perp^2 \rangle_{n,l,m} = \frac{m^2(Z\alpha)^2}{n^2} - \langle \hat{\mathbf{p}}_\perp^2 \rangle_{n,l,m} = \frac{m^2(Z\alpha)^2}{n^2}\left(1 - \frac{l(l+1)}{n(l+1/2)}\right). \tag{26}$$

Note that for $l = 0$, we have $\sqrt{\langle \hat{p}_r^2 \rangle_{n,l,m}} \simeq m(Z\alpha)/n = mv_n$, further justifying the interpretation of v_n as the velocity of the electron in the n-th energy level. However, for $l \neq 0$, the radial momentum is reduced. In fact, for maximum $l = n-1$, we have

$$\langle \hat{p}_r^2 \rangle_{n,l,m}^{\min} \simeq \frac{m^2(Z\alpha)^2}{n^2}\left(1 - \frac{n(n-1)}{n(n-1/2)}\right) \simeq \frac{m^2(Z\alpha)^2}{n^2}\frac{1}{2n} = \frac{m^2 v_n^2}{2n} \quad \text{for large n and } l = n-1, \tag{27}$$

which resembles the classical picture of an electron in a circular orbit, where the radial momentum is zero and the kinetic energy is purely transverse.

The discussion above suggests we can interpret $v_n = (Z\alpha)/n$ as the velocity of the electron in the n-th energy level. In this way, it is easy to remember the energy levels of the single-electron atom,

$$\boxed{E_n^{(0)} = -\frac{mv_n^2}{2}, \quad v_n = \frac{Z\alpha}{n}.} \tag{28}$$

This is just the classical kinetic energy of a particle with mass m and velocity v_n.

Let us put in some numbers:

$$
\begin{aligned}
E_n^{(0)} &\simeq 13.6\ \text{eV}\left(\frac{Z}{n}\right)^2, && (29)\\
a_0 &\simeq 0.5\times 10^{-10}\ \text{m}\times\left(\frac{1}{Z}\right), && (30)\\
\frac{1}{a_0} &\simeq 3.8\ \text{keV}\times Z, && (31)\\
v_n &\simeq 7\times 10^{-3}\left(\frac{Z}{n}\right), && (32)\\
t_{\text{electron}} = \frac{a_0}{v_n} &\simeq 2.4\times 10^{-8}\ \text{ns}\times\frac{n}{Z^2} && (33)
\end{aligned}
$$

In the last equation we used t_{electron} as the characteristic time scale of the system. These are the typical dimensionful scales in atomic physics. Note that the energies involved are far smaller than the electron mass, so we can safely work in the non-relativistic limit (that is, $v_n \ll 1$).

Symmetries: The degeneracy of the energy levels in our single electron atom is a consequence of the symmetries of the Coulomb potential. The first is spherical symmetry, or the invariance under the group of rotations $SO(3)$. This leads to the conservation of the orbital angular momentum $\mathbf{L}$ and the associated quantum numbers ℓ and m. There is, however, an additional hidden symmetry related to the conservation of the Runge-Lenz vector ((see, e.g., S. Weinberg, "*The* $SO(4)$ *Symmetry of the Hydrogen Atom*"). This additional symmetry is responsible for the degeneracy between states with different ℓ but the same n. It can be seen as an enhancement of the $SO(3)$ symmetry to $SO(4)$, which is the group of rotations in four dimensions!

This is highly significant: it is telling us that the motion of the electron in the Coulomb potential is invariant under rotations in four-dimensional space; admitedly, I do not have a good intuition as to what this fourth dimension means, but we can write the conserved vector:

$$
\mathbf{A} = \frac{1}{2m}(\mathbf{p}\times\mathbf{L} - \mathbf{L}\times\mathbf{p}) - Z\alpha\frac{\mathbf{r}}{r}. \qquad (34)
$$

In Keplerian mechanics, this vector points along the direction of the major axis, and so, it makes sense that if we have closed orbits (i.e., orbits that are not precessing), this vector, or at least its direction, should be conserved. This vector points along the direction of the major axis of the electron's orbit and its magnitude is related to the eccentricity of the orbit.

Let us compute the expectation value of $\mathbf{A}^2 = \mathbf{A}\cdot\mathbf{A}$ in the state $|n,\ell,m\rangle$. I will just give the result here:

$$
\langle\mathbf{A}^2\rangle_{n,\ell,m} = \frac{Z^2\alpha^2}{n^2}\left(1 - \frac{\ell(\ell+1)}{n^2}\right). \qquad (35)
$$

(Note the resemblence with the expression for the radial momentum above.) From Keplerian mechanics, the eccentricity of the orbit is given by $e = |\mathbf{A}|/(Z\alpha)$, so

$$
e_{n,\ell} = \sqrt{1 - \frac{\ell(\ell+1)}{n^2}}. \qquad (36)
$$

This means that the states with $\ell = 0$ are the most eccentric (we can imagine the electron zipping back and forth along a radial line), while the states with $\ell = n-1$ are the least eccentric, resembling circular orbits.